

A non-invasive tool for assessing pathogen prevalence in koala (*Phascolarctos cinereus*) populations: detection of *Chlamydia pecorum* and koala retrovirus (KoRV) DNA in genetic material sourced from scats

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Received: 1 March 2016 / Accepted: 5 July 2016 / Published online: 8 July 2016
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Abstract Pathogenic diseases may threaten the viability of wild animal populations, especially when already vulnerable. The mitigation of risks associated with pathogenic infections in populations is an important factor in conservation strategies. Koalas are of conservation concern across the north of their range and are affected by two main pathogens; *Chlamydia pecorum* and the koala retrovirus (KoRV). This study tested whether DNA from *C. pecorum* and KoRV could be detected in genetic material isolated from koala scats. Detection of *C. pecorum* in scat isolated DNA samples was compared with results obtained from urogenital swabs collected from the same individuals as part of an independent study. The ability to detect KoRV in scats from both northern and southern regions of the koala's range was also assessed. There was a high level of concordance (5/6) between the detection of *C. pecorum* in DNA isolated from scats and urogenital swabs from the same individual. In positive samples, *C. pecorum ompA* genotypes were identical between DNA from scats and urogenital swabs in two out of three cases. In samples from the south of the koala's range, KoRV copy number was higher in DNA isolated from scats compared to DNA isolated from ear tissue, potentially indicating the detection of horizontally acquired infections. Our results demonstrate

the ability to detect *C. pecorum* and KoRV in DNA isolated from koala scats. This method will be useful for studying the prevalence, transmission and impact of these pathogens in wild populations which may subsequently inform conservation management strategies.

Keywords Non-invasive pathogen detection · Faecal samples · *Chlamydia pecorum* · Koala retrovirus (KoRV) · Wildlife health monitoring · Wildlife conservation · Quantitative PCR

Introduction

Pathogens infecting wild animal populations can affect the health of individuals and may lead to reduced fitness and mortality. Pathogenic diseases can reduce species viability and have been implicated in wild population declines (Tompkins et al. 2015). Consequently, pathogens may have a negative impact on biodiversity and their role as a potential driver of extinctions is becoming increasingly apparent (Smith et al. 2009). Greater data collection may facilitate characterisation and understanding of pathogen prevalence, transmission and impacts, and has the potential to inform and enhance management and conservation strategies.

Endemic to Australia, the koala (*Phascolarctos cinereus*) is an arboreal marsupial inhabiting the country's east coast, from Queensland to South Australia (Fig. 1), that has suffered dramatic population declines over the past two decades (Department of the Environment 2015). While the causes of decline are likely multifactorial, two pathogens have potential contributory roles: the obligate intracellular bacterium, *Chlamydia pecorum*, and the koala retrovirus (KoRV), both of which infect koalas throughout their

Electronic supplementary material The online version of this article (doi:10.1007/s12686-016-0574-3) contains supplementary material, which is available to authorized users.

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Fig. 1 Koala distribution (shown by grey shading) across Australia adapted from Department of the Environment (2015)

range. *C. pecorum* commonly infects the eyes and/or urogenital tract of koalas (Markey et al. 2007). Ocular infection with *C. pecorum* may lead to severe keratoconjunctivitis (Cockram and Jackson 1981), while urogenital infections are known to result in cystitis and reproductive disease, which may lead to decreased fecundity or sterility (Martin and Handasyde 1990a; Obendorf and Handasyde 1990). A second species of *Chlamydia*, *C. pneumoniae*, also infects koalas but is usually associated with mild symptoms of disease (Jackson et al. 1999). There is evidence that different strains of *C. pecorum* have varying pathogenic potential (Mohamad et al. 2008, 2014) and that apparently asymptomatic chlamydial infections may have hidden impacts. In cattle, for example, reduced calf growth rates have been associated with asymptomatic *C. pecorum* infection (Poudel et al. 2012). The potential impacts of *C. pecorum* on individual koalas and koala populations highlight the need for better characterisation and understanding of the prevalence of *C. pecorum* infection.

KoRV is a retrovirus that is considered to be endogenous in northern koala populations and in the process of becoming endogenous in southern populations (Simmons et al. 2012). Although numerous strains of KoRV have been reported to date (Xu et al. 2013, 2015), KoRV-A is most commonly detected. Unlike most known endogenous viruses, proviral KoRV (viral DNA inserted into the host genome) is capable of producing active virus (Tarlinton et al. 2005); KoRV may therefore be transmitted vertically, from parent to offspring, or horizontally via contact transmission. Modes of KoRV horizontal transmission are not clear, however KoRVs closest known relative, the gibbon ape leukaemia virus (GALV) can also be transmitted both vertically, in the germline, and horizontally, via exposure to faeces or urine (Kawakami et al. 1977, 1978). Though not definitive, links have been made between KoRV infection and the incidence of lymphoma and leukaemia in koalas, as well as with clinical chlamydial infections (Tarlinton et al. 2005; Xu et al. 2013). In the north of the koala's range (from Port Macquarie northwards; Fig. 1), 100 % of koalas tested were found to be infected with KoRV. However, in the south-eastern state of Victoria, there is evidence that endogenous KoRV is less common, with virus integrated only into infected cells (exogenous infection) and a significant number of Victorian koalas tested are found to be KoRV negative (Simmons et al. 2012).

Knowledge of the epidemiology of *C. pecorum* and KoRV across the koala's range could identify pathogen free populations and/or populations at high risk of increased mortality due to infection. Detecting pathogenic organisms in wild animals is often challenging, as sampling typically involves animal capture which can be difficult, especially where the species is elusive or when population densities are low. Once located, site factors and an animal's position, often, in the case of koalas, in the tops of trees exceeding 30 metres in height within rugged terrain, may also limit accessibility and make capture difficult or impossible. Animal capture is usually followed by invasive procedures such as anaesthesia and the collection of urogenital or ocular swabs, tissue biopsies or blood extraction. These methods require veterinarians and/or specialist expertise, making widespread sampling costly and time consuming. Due to difficulties in sampling wild animals, the prevalence of pathogens is often determined using samples of convenience such as deceased individuals (e.g. roadkill) or those admitted to veterinary clinics or wildlife shelters. Though a subset of the wild population, the prevalence of infectious agents may be greater in the portion of the population that has met with misadventure or is unwell (entering shelters or clinics) compared to the overall wild population. Therefore, this type of sampling has the potential to lead to a biased result, which may not be

indicative of the true prevalence of the pathogen within the wider population.

As in other marsupials, both male and female koalas have a cloaca, a common external opening to the urogenital and gastrointestinal tracts (Archer et al. 1987). It therefore seems probable that dead *Chlamydia* infected cells and shed infectious particles (elementary bodies) may be detected in genetic material sourced from koala scats. The ability to carry out microsatellite genotyping using koala DNA isolated from scats (Wedrowicz et al. 2013) suggests that it should also be possible to detect KoRV in infected epithelial cells. In koalas originating from Queensland, the ability to detect endogenous KoRV (where all cells will contain at least one KoRV copy) in DNA isolated from koala scats has been demonstrated by Miyazawa et al. (2011). However, the ability to detect exogenous KoRV infections (in scat isolated DNA), where all somatic cells are not infected, is less certain. Given evidence that a large proportion of Victorian koalas may only carry horizontally acquired, exogenous KoRV infections (Simmons et al. 2012), this is important to determine.

The detection of koala pathogens in non-invasively sourced DNA could provide a viable alternative to invasive sampling. The ability to detect *C. pecorum* and KoRV in DNA isolated from koala scats would be a valuable tool with the potential to facilitate and expedite the collection of prevalence data for these infections across the koala's range. The objective of this study, therefore, is to determine whether *C. pecorum* and KoRV-A DNA can be detected in genetic material non-invasively sourced from koala scats, providing a tool that can assess the presence of these pathogens in wild koala populations.

Materials and methods

Chlamydia pecorum

Sample collection and DNA isolation

Scat samples were collected from six koalas in the Strzelecki Ranges, Victoria during March 2015. The six individual koalas from which scats were collected were also captured as part of an independent study carried out by the NSW Office of Environment and Heritage (Allen 2015). The purpose of the study carried out by Allen (2015) was to determine the prevalence of chlamydial infections in the Strzelecki Ranges koala population by capturing and collecting ocular and urogenital swabs from koalas. Results obtained from DNA isolated from urogenital swabs (six koalas: K01–K06) collected for the Allen (2015) study allowed direct comparison of *C. pecorum* presence detected by both methods. Scats located on the ground beneath

each koala captured by Allen (2015) were collected for the isolation of DNA according to the methods described by Wedrowicz et al. (2013). To assess detection reproducibility in DNA isolated from scats from the same individual, DNA was isolated from three different scats collected beneath each koala, resulting in three DNA isolates per individual. The concentration of total DNA isolated from scats was determined using the Qubit® 2.0 Fluorometer following the manufacturer's instructions.

Presence/absence assay for C. pecorum

Real time assays for *C. pecorum* targeting a 76 bp region of the *ompA* gene were carried out as per Pantchev et al. (2010). Reactions also incorporated TaqMan® Exogenous Internal Positive Control (IPC) Reagents in order to differentiate failed reactions, due to PCR inhibition, from true negative results. Amplification was carried out using the Applied Biosystems Step One Plus instrument using a presence absence protocol. PCR product using primers amplifying 1140 bp of the *ompA* gene (details below: *ompA* gene sequencing) was used to prepare 1:10 serial dilutions for standards ranging from 8×10^0 – 8×10^6 copies per reaction. Standards were used to assess reaction sensitivity including an IPC, as well as to estimate the number of chlamydial DNA copies in different samples. A standard PCR to determine the gender of the sampled individual was also carried out using DNA isolated from scats as outlined in Wedrowicz et al. (2013).

ompA gene sequencing

The presence of *C. pecorum* in samples which tested positive via real time PCR was confirmed by amplification of 1140 bp of the *ompA* gene. Amplification of *ompA* was carried out using BIO-X-ACT™ Short DNA Polymerase (Bioline). Reactions consisted of 1× OptiBuffer, 1× Hi-Spec additive, 2 mM MgCl₂, 0.5 mM each dNTP, 1 unit of BIO-X-ACT Short DNA Polymerase and 0.25 μM of each primer (*ompA*for and *ompA*rev; Kollipara et al. 2013) made up to 20 μL with water. DNA was initially denatured for 5 min at 95 °C, followed by 40 cycles of 95 °C for 30 s, 54 °C for 30 s and 72 °C for 1 min each and ending with a single final extension of 7 min at 72 °C. PCR products were purified using the Wizard® SV Gel and PCR Clean-Up System (Promega) and sequencing was carried out by the Australian Genome Research Facility (AGRF), Melbourne, Victoria. Resulting sequences were trimmed using Sequence Scanner V2 (Applied Biosystems) and nucleotide BLAST (NCBI 2016) used to determine the similarity of sequences to those previously reported. Genotypes were classified according to Kollipara et al. (2013), where *ompA* sequences with less than 1 % nucleotide differences to

previously reported sequences were designated by the same genotype letter followed by a prime symbol and a number to differentiate multiple variants with minimal base pair differences.

Comparison of scat and urogenital results

Results of *C. pecorum* presence in DNA isolated from scats were compared to *C. pecorum* presence in urogenital swabs obtained by Allen (2015). Results reported by Allen (2015) were determined independently by the Koala Health Hub (KHH) at The University of Sydney, New South Wales, Australia. DNA isolated from swabs testing positive for *C. pecorum* were provided for this study by the KHH allowing amplification and sequencing of the *ompA* gene to be carried out for swab isolated DNA as described above.

Koala retrovirus (KoRV)

Sample collection and DNA isolation

Koala scat samples were collected from wild individuals in the Strzelecki Ranges, South Gippsland ($n = 17$), Raymond Island ($n = 2$) and from koalas at the Koala Conservation Centre ($n = 6$) on Phillip Island, all in Victoria (total VIC samples; $n = 25$) (Fig. 1). Koala scats were also obtained from the Port Macquarie Koala Hospital in New South Wales (NSW, $n = 15$). Scats were collected and DNA isolated according to the methods outlined in Wedrowicz et al. (2013). To compare the results from scat samples with those from a commonly utilised invasive DNA source, DNA was also isolated from ear tissue samples ($n = 50$) from deceased individuals, collected by staff at the Southern Ash Wildlife Shelter (SAWS), Rawson, Victoria. Approximately 3 mm square was excised from ear samples and cut into small pieces. DNA was then isolated using the DNeasy® Blood & Tissue Kit (Qiagen) following the manufacturer's protocol.

KoRV-A PCR

Standard PCRs utilised KoRV-A specific primers published in Xu et al. (2013) alongside koala β -actin primers (Markey et al. 2007) to confirm the presence of koala DNA. Standard PCRs for KoRV-A were made up of 5 μ L GoTaq® Green Master Mix (Promega), 1.0 μ M of each KoRV-A primer, 0.1 μ g bovine serum albumin (BSA), 1 μ L of DNA template and water to give a final volume of 10 μ L. PCR cycling parameters were 95 °C for 2 min followed by 35 cycles of 95 °C for 30 s, 64 °C for 30 s and 72 °C for 30 s followed by a final extension of 72 °C for 5 min. Standard PCR for koala β -actin consisted of 5 μ L GoTaq® Green Master Mix, 1.0 μ M of each KoRV-A

primer, 0.1 μ g BSA, 1 μ L of DNA template and water to bring the final volume to 10 μ L. PCR cycling parameters were 95 °C for 2 min followed by 35 cycles of 95 °C for 30 s, 60 °C for 30 s and 72 °C for 30 s and finished with a final extension of 72 °C for 5 min. To ensure the lower sensitivity of standard PCR assays was not resulting in an increased number of false negatives, real time PCR assays, using the same primers specific for KoRV-A (Xu et al. 2013), were carried out by the KHH on the same set of samples (VIC and NSW, $n = 40$) tested by standard PCR to independently validate results. Real time PCRs were carried out on both undiluted and 1:10 dilutions of DNA isolates.

KoRV-A copy number

DNA from tissues ($n = 50$) and additional scat samples ($n = 117$) collected in the South Gippsland region were also screened for KoRV (data not shown). Using a subset of randomly selected positive samples, KoRV copy number was compared between Victorian scat samples (VIC, $n = 13$) and ear tissue samples (TISS, $n = 13$) along with scat samples collected from New South Wales (NSW, $n = 8$). This was carried out using real time PCR with KoRV-A primers (Xu et al. 2013) and β -actin primers (Markey et al. 2007) to standardise copy number. Standards were produced for the KoRV-A and β -actin assays by amplifying the two targets using standard PCR. KoRV-A and β -actin PCR product was then purified using the Wizard® SV Gel and PCR Clean-Up System (Promega). Serial 1:10 dilutions ranging approximately from 10^0 to 10^6 copies of KoRV-A or β -actin PCR product were then used as standards for the estimation of copy number. Both PCRs were carried out using 10 μ L of SYBR® Green Master Mix (Applied Biosystems), 0.25 μ M of each primer and 2 μ L of DNA template made up to a total of 20 μ L with water and cycling parameters consisting of 50 °C for 2 min (UDG activation), 95 °C for 2 min (polymerase activation) followed by 50 cycles of 95 °C for 3 s and 60 °C for 30 s. A melt curve was produced to check for non-specific products. All standards, controls and samples were tested in duplicate.

KoRV-A sequencing

A subset of PCRs positive for KoRV ($n = 6$) were sequenced to confirm the detection of KoRV-A. Primers were designed using published KoRV-A sequence (Genbank AF151794.2; Hanger et al. 2000) and Primer-BLAST (Ye et al. 2012) to target a 1115 bp region of the KoRV *env* gene; primers were designated KoRV-*env*1-F (5'-AGACG GGAAGTGTCTGTTTGG-3') and KoRV-*env*1-R (5'-GGG GGTGAGGCCAGAATTAC-3'). Positive samples were

amplified using 10 µL GoTaq® Green Master Mix, 0.5 µM of each primer and 0.1 µg BSA made up to a final volume of 20 µL. PCR products were purified using the ISOLATE PCR and Gel Kit (Bioline) and sequencing was carried out by AGRF.

Results

Chlamydia pecorum

Assay sensitivity and reproducibility

The detection limit for the *C. pecorum* assay, with the addition of an internal positive control, was consistent with the limits reported by Pantchev et al. (2010) at eight copies of chlamydial DNA per PCR (Online resource 1, Fig. S1). *C. pecorum* DNA was detected in all three scats from each positive individual indicating that, when present, *C. pecorum* is consistently found in DNA isolated from koala scats. The total number of *C. pecorum* copies detected in DNA isolated from urogenital swabs was much higher compared to scats (Table 1). Results obtained from scat or

urogenital swab isolated DNA were generally concordant (Table 2), but, one of the six scat DNA isolates was positive for *C. pecorum* while the urogenital swab was negative (K06).

C. pecorum copy number and DNA concentration

DNA isolated from scats ranged in concentration from 1.4 ng/µL up to 42 ng/µL (Table 1; Fig. 2). The estimated copy number of *C. pecorum* increased with increasing DNA concentration (Fig. 2). Different scats from the same individual were consistently either positive or negative, though there was variation in the number of *C. pecorum* copies detected for independent scats from the same individual (Table 1). Of the fifteen DNA samples isolated from scats positive for *C. pecorum*, twelve had less than 10 *C. pecorum* copies per ng of DNA isolated while the remaining three samples had somewhat higher copy numbers; 15, 37 and 74 *C. pecorum* copies per ng of DNA (Table 1). Compared to DNA isolated from scats for each individual, *C. pecorum* copy number was on average, three (K02), ten (K01 and K04), and 3000 (K05) times greater in DNA isolated from swabs.

Table 1 *C. pecorum* copy number and DNA concentration for each isolate tested in this study

Koala ID	Sample	Copy number	DNA concentration (ng/µL)	Copy number per ng of total DNA
K01	Scat 1	1.71	1.42	1.21
	Scat 2	8.86	1.91	4.63
	Scat 3	14.4	1.60	9.00
	Swab	84.8	nt	nt
K02	Scat 1	155	2.10	73.6
	Scat 2	21.9	3.92	5.57
	Scat 3	8.57	2.54	3.38
	Swab	203	nt	nt
K03	Scat 1	NA	14.0	NA
	Scat 2	NA	7.88	NA
	Scat 3	NA	6.46	NA
	Swab	Negative	nt	NA
K04	Scat 1	624	41.6	15.0
	Scat 2	241	6.52	37.0
	Scat 3	358	42.2	8.49
	Swab	3840	nt	nt
K05	Scat 1	16.7	6.62	2.53
	Scat 2	34.2	9.10	3.76
	Scat 3	23.4	10.9	2.16
	Swab	74,800	nt	nt
K06	Scat 1	12.9	1.90	6.81
	Scat 2	8.91	5.46	1.63
	Scat 3	7.21	4.38	1.65
	Swab	Negative	nt	NA

NA not applicable, nt not tested

Table 2 Summary of results for PCRs targeting *C. pecorum* in DNA isolated from koala scats with comparison to results reported by Allen (2015) from swab samples

Koala ID (Allen 2015 ID)	DNA isolated from scats (this study)					DNA isolated from urogenital swabs (Allen 2015)		
	XY PCR	Real time assay	<i>C. pecorum</i> copy number per ng (mean $\pm$ SE)	<i>ompA</i> genotype	Sequence similarities	Gender	Real time assay	<i>ompA</i> genotype
K01 (STRZ009)	Female	Positive	4.95 $\pm$ 2.25	F'3	99 % identity to KU214246.1	Female	Positive	F'3 (99 % identity to KF150135.1)
K02 (STRZ010)	Female	Positive	27.5 $\pm$ 23.0	F'3	99 % identity to KU214246.1	Female	Positive	M (100 % identity to KU214247.1)
K03 (STRZ011)	Female	Negative	NA	Negative	NA	Female	Inconclusive ^b	NA
K04 (STRZ012)	Female	Positive	20.1 $\pm$ 8.60	C'1	99 % identity to KU214245.1	Female	Positive	C'1 (99 % identity to KU214245.1)
K05 (STRZ013)	Male	Positive	2.82 $\pm$ 0.485	–	Sequence unreadable ^a	Male	Positive	F (100 % identity to KF150135.1)
K06 (STRZ014)	Female	Positive	3.36 $\pm$ 1.72	F'3	99 % identity to KU214246.1	Female	Negative ^c	NA

ompA genotypes and comparisons are based on those reported by Kollipara et al. (2013) and Legione et al. (2016)

NA not applicable, SE standard error of the mean

^a Unreadable sequence obtained, possibly due to the presence of multiple strains

^b Sample was positive for the *C. pecorum* specific assay (but with low copy number, $C_T = 38$), but negative using the *Chlamydia* genus and *C. pneumoniae* assays (Allen 2015)

^c Sample was negative using species specific (both *C. pecorum* and *C. pneumoniae*) assays and positive (but with low copy number, $C_T = 37$) using a *Chlamydia* genus PCR (Allen 2015)

ompA genotypes detected using DNA isolated from scats and swabs

Of the five *Chlamydia* positive samples detected using DNA isolated from scats, four were successfully sequenced at *ompA* with two different genotypes being detected (Table 2). Referring to the types reported by Kollipara et al. (2013), genotypes were most closely related to *ompA* types F (K01, K02 and K06; F'3, $n = 3$, two bases difference to genotype F, both non-synonymous) and C (K04; C'1, $n = 1$, one base difference, synonymous).

Each of the four positive urogenital swab samples produced different *ompA* sequences to one another (Table 2). These included F'3 (K01) and C'1 (K04), both of which matched the sequences obtained from counterpart scat samples. An *ompA* sequence 100 % identical to genotype F described by Kollipara et al. (2013) was detected in the swab sample of koala K05, but the *ompA* sequence obtained for K05 using scat isolated DNA could not be read, possibly due to the presence of multiple strains. The remaining swab sample (K02) was identified as genotype M (Legione et al. 2016) and differed to the corresponding *ompA* sequence from the paired scat isolated DNA (F'3). The *ompA* sequence (genotype M) obtained from swab DNA for individual K02 was noted to have some level

background sequence while all other sequences were completely free of any background noise.

Koala retrovirus (KoRV)

Comparison between standard and real time PCR results

The DNA concentration isolated from the 40 scat samples tested for KoRV presence ranged from undetectable (<0.10 ng/ μ L, $n = 3$ NSW samples) to 41 ng/ μ L with a mean of 4.2 ng/ μ L and median of 1.2 ng/ μ L (Online resource 1: Table S2). Standard PCR results (undiluted template) were identical to the real time PCR results (undiluted template), except for three (12.5 %) South Gippsland positives which tested negative by real time PCR when isolates were diluted 1:10 (Online resource 1: Table S2). KoRV-A was detected in 24 of the 40 samples tested (60 %). Using both standard and real time PCR all six Phillip Island samples tested in this study were negative for KoRV while all 15 NSW samples tested positive. As expected, β -actin and KoRV-A cycle thresholds decreased with increasing amounts of total DNA concentration obtained from koala scats (Fig. 3), indicating greater amounts of koala and KoRV DNA present in DNA isolates with higher DNA concentration.

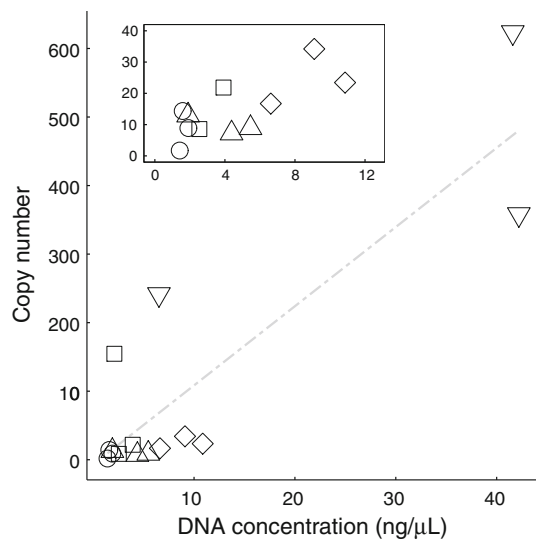


Fig. 2 Plot showing the relationship between *C. pecorum* copy number and DNA concentration in DNA isolated from scat samples testing positive for *C. pecorum* (K01, K02, K04, K05 and K06). Symbols represent the independent scats from the same individual: K01 (open circle), K02 (open square), K04 (open inverted triangle), K05 (open diamond) and K06 (open triangle). The overall regression equation was *C. pecorum* copy number = $-7.0 + 11.6 \times \text{DNA concentration (ng/}\mu\text{L)}$ ($R^2 = 74.3$, $p < 0.0005$). The cluster of data points with DNA concentration below 12.5 ng/μL and copy number below 40 *C. pecorum* copies per genome is shown in greater detail in the inset at the top left of the figure

Confirmation of KoRV-A presence by sequencing

A 1115 bp segment of the KoRV *env* gene was sequenced for a subset of positive samples to confirm the detection of KoRV DNA. Sequences of samples from NSW koalas ($n = 2$) were 100 % identical to KoRV-A sequence (AF151794.2; Hanger et al. 2000). All samples from Victorian koalas that were sequenced ($n = 4$) had six identical nucleotide base changes in the *env* segment. The six polymorphisms in the Victorian samples were all synonymous, not having an effect on the amino acid sequence of the *env* gene.

Differences in KoRV-A copy number between sample groups

Quantitative PCR revealed significant differences in copy number between scat samples from NSW and Victoria. On average, more than four KoRV-A copies per genomic unit were detected in scats sampled in NSW, whereas less than one copy per genomic unit was detected in Victorian scat samples (Fig. 3; $p = 0.01$). This is concordant with results reported by Simmons et al. (2012) further supporting the hypothesis that most infections in Victoria are exogenous, as endogenous proviral DNA would be expected to be

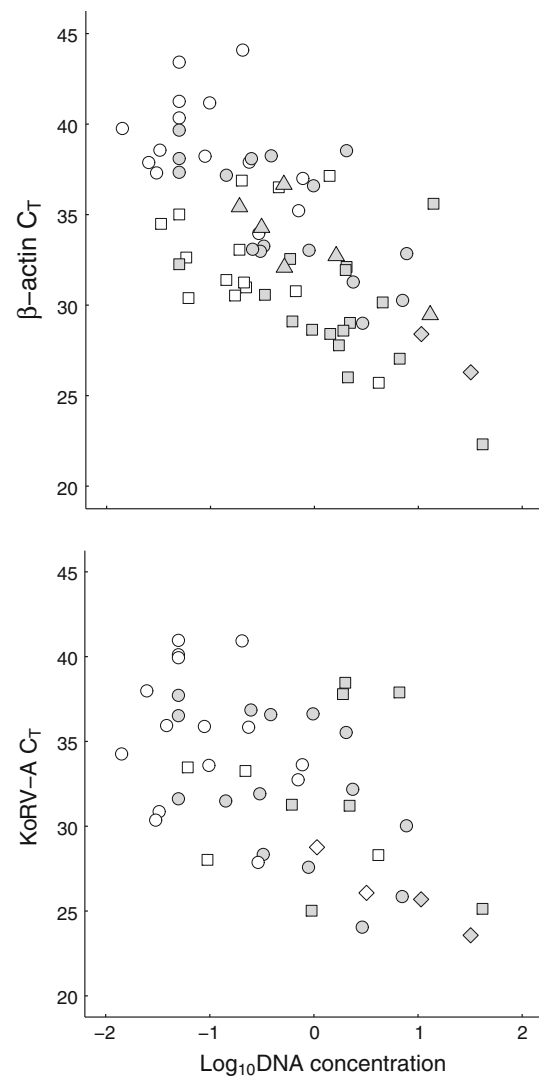


Fig. 3 The relationship between cycle threshold (C_T) and DNA concentration for positive β -actin and KoRV-A real time PCRs. Symbols indicate samples from NSW (open circle), South Gippsland (open square), Raymond Island (open diamond) and Phillip Island (open triangle). Unshaded symbols denote DNA isolates that were diluted 1:10 while shaded symbols indicate undiluted samples. The regression fit for the β -actin PCR is $C_T = 32.5 - 3.7 \log_{10}\text{DNA}$ ($R^2 = 42\%$, $p = 1.1 \times 10^{-9}$) while the KoRV-A PCR had a regression fit of $C_T = 31.6 - 2.9 \log_{10}\text{DNA}$ ($R^2 = 26\%$, $p = 1.9 \times 10^{-4}$)

present in every cell if acquired vertically via the germline. Simmons et al. (2012) showed that KoRV is endogenous in northern koala populations, and is inserted into the genome multiple times, with an average of 165 KoRV copies per cell reported for Queensland individuals; the mean of four KoRV copies per genomic unit value reported for NSW koalas in this study is substantially lower. KoRV-A copy number also differed significantly between scat and ear tissue samples collected in South Gippsland, with scats having on average, 30 times as many KoRV-A copies per

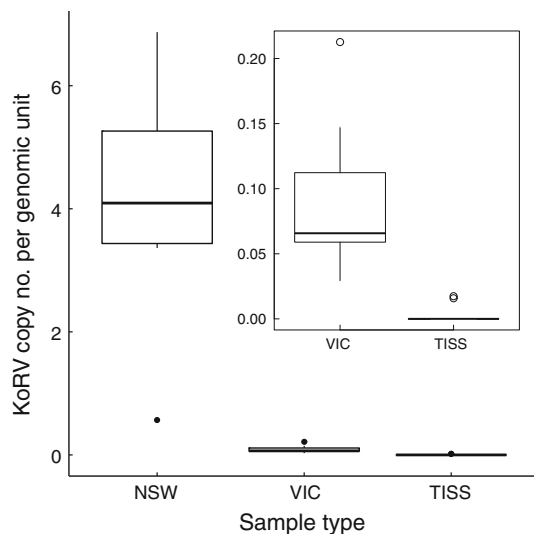


Fig. 4 Differences in KoRV-A copy number per genomic unit (β -actin) between sample groups. Scat samples from New South Wales (NSW, $n = 8$) had a mean of 4.15 ± 0.66 KoRV-A copies per genomic unit, scat samples from Victoria (VIC, $n = 13$) had a mean of 0.087 ± 0.014 copies per genomic unit and tissue samples from Victoria (TISS, $n = 13$) had 0.003 ± 0.002 KoRV-A copies per genomic unit. All groups differed significantly from one another ($p \leq 0.0005$). The inset shows the data for Victorian scat (VIC, $n = 13$) and tissue samples (TISS, $n = 13$) on a smaller scale

genomic unit than tissue samples (Fig. 4; $p < 0.0005$). KoRV-A in DNA isolated from ear tissue averaged 0.003 ± 0.002 copies per genomic unit while DNA isolated from scats had a mean of 0.09 ± 0.01 KoRV-A copies per genomic unit.

Discussion

Pathogenic organisms can impact wild animal populations by reducing animal health, longevity and the ability to successfully reproduce. Impacts such as these may result in decreased population size causing losses in genetic diversity (Frankham et al. 2012) which, together with other threatening processes such as habitat loss and fragmentation, may greatly increase the risk of inbreeding, deleterious genes and extirpation. Given declines in koala numbers and the 2012 listing of northern koala populations as vulnerable under the *Environmental Protection and Biodiversity Conservation Act 1999* (Department of the Environment 2015) it is imperative that the prevalence of *C. pecorum* and KoRV within the koala population is determined. Obtaining such data via swabs however, can be costly and time consuming, especially for species such as the koala, which are difficult to capture. DNA isolated from a single scat can provide information on the presence and characteristics of *C. pecorum* and KoRV. Both *C.*

pecorum and KoRV were detected in DNA isolated from koala scats. The non-invasive collection of samples therefore has potential to greatly increase sample size in studies of koala pathogens, without the requirements of capturing, handling or even observing an animal. This allows for the collection of more samples from wider and more diverse geographic areas, increasing both the spatial distribution and number of samples obtained for studies of pathogens in wildlife. Such data is critical for effective species management, especially in the case of inter-population translocations or releases proposed by wildlife managers or carers.

In Australia, the translocation of wild animals is common, with 54 species ($\sim 14\%$) of Australian mammals having at least one documented translocation (Short 2010). Additional undocumented movements of animals carried out by wildlife carers releasing individuals after rehabilitation are also commonplace in Australia (Guy and Banks 2012; Reid 2014). Translocations are often carried out for conservation reasons, however the risk of transferring pathogens between endemic and introduced individuals is an important consideration that requires prior assessment (Leighton 2002) and post translocation surveillance (Mathews et al. 2006). Severe *Chlamydia* infections are more likely in individual koalas not previously exposed to these bacterial infections (Martin and Handasyde 1990b). The inadvertent introduction of *C. pecorum* to populations that have historically been free of infection, therefore, has the potential to result in devastating consequences. The effects of exposure to novel strains are also important considerations due to possible differences in strain pathogenicity (Mohamad et al. 2008, 2014). Although the pathogenic effects of KoRV are not entirely clear at present, preventing transmission of the virus to KoRV free populations is important. Obtaining baseline data on KoRV prevalence in wild populations will enable monitoring of infection rates and future changes that may occur.

C. pecorum has also been found to infect other Australian marsupials such as the greater glider (*Petauroides volans*), the short-eared brushtail possum (*Trichosurus caninus*) and the western barred bandicoot (*Perameles bougainville*) (Bodetti et al. 2003). Other than the koala, limited research into the effects of this pathogen on marsupial populations has been carried out. Given that urogenital infections with *Chlamydia* may be subclinical, yet affect reproduction, the presence of this bacteria in marsupial populations may potentially be an under recognised cause of population declines. The use of scat sourced DNA could provide a simple and useful method for screening and characterising *Chlamydia* infections in a range of marsupial species. Similarly, a retrovirus similar to KoRV and GALV has been detected in an Australian rodent, *Melomys burtoni* (Simmons et al. 2014). Using DNA

sourced non-invasively from scats, sampling power and the ability to identify other potential retroviral hosts and/or cross species transmissions may be greatly increased. This method could therefore also be of value to the study of these bacterial and viral pathogens in other Australian marsupials. Screening for potential pathogens, utilising non-invasive genetic sampling techniques as described here, may be a useful addition to studies of other mammals.

This study demonstrates that *C. pecorum* strains, also found to cause urogenital infection in koalas (Kollipara et al. 2013; Legione et al. 2016), can be detected in DNA isolated from koala scats. It is important to recognise that infection with *C. pecorum* does not mean that an individual will have symptoms of chlamydiosis. The presence of *C. pecorum* does, however, provide an indication that the individual carries the pathogen and is probably capable of transmitting the bacteria to other individuals. Conversely, a negative result does not necessarily show that the individual is free of infection, since latent infections or low level shedding of infectious particles could result in too few *C. pecorum* particles present on koala scats for detection.

Results of the *C. pecorum* assay were largely concordant between scat and swab samples, although some differences were also noted. One individual (K06) tested positive for *C. pecorum* in DNA isolated from scats and negative using DNA isolated from the urogenital swab. This may indicate that insufficient biological material was obtained during swabbing or that infected cells detected in scat DNA were located at other reported sites of chlamydial infection such as the gastrointestinal tract (Burach et al. 2014). Another explanation is that scats were from another koala having visited the same tree rather than the koala captured; genotyping of both the scat and swab DNA isolates would be required to confirm or exclude this possibility. The *ompA* genotype obtained for one individual (K02) differed between scat and swab samples; the scat sample providing the F'3 genotype and the swab sample the M genotype. The M genotype sequence obtained from this swab was however the only sequence with a substantial amount of background, potentially indicating the presence of multiple strains or alternatively, sample contamination. Further investigation is needed to determine whether differences in the *ompA* type detected using different sample types from the same individual is due to the collection of scats from unintended individuals, as described above, or to multiple strain infections, potentially located at different anatomical sites (Burach et al. 2014). For example, infections of the gastrointestinal tract may be present in scat samples, but absent from urogenital swab samples.

Compared to DNA isolated from swabs, the number of *C. pecorum* copies detected in scat DNA was much lower and the number of estimated *C. pecorum* copies varied

between DNA samples isolated from independent scats from the same individual. Although positive samples were consistently positive across the three separate scats, it is possible that the bacteria may not be detected when the rate of shedding of infectious particles is low. Copy number variability was at least, in part, related to the concentration of DNA obtained from scats. Further work to determine the minimum quantity of total DNA needed for reliable detection of *C. pecorum* in DNA isolated from scats, from individuals with various infectious loads, would be appropriate. Testing multiple independent scats from the same individual may increase confidence in results obtained. The only male koala included in this study (individual K05) was estimated to have a *C. pecorum* copy number that was between 2200 and 4500 times lower in DNA sourced from scats compared to DNA sourced from swabs, while the females in the study had copy numbers in scats that were between one and fifty times lower. Differences in the anatomy of female and male koalas could potentially affect the ability to detect *C. pecorum* in males and female scats; this requires further investigation.

Compared to Victorian tissue samples, significantly greater amounts of KoRV-A were detected in DNA isolated from Victorian scat samples (Fig. 4). This difference suggests the possible detection of infection acquired horizontally (exogenous infection) in Victorian koalas. All northern NSW samples in this study tested positive for KoRV, while no samples were KoRV positive in the Phillip Island group of samples. This is consistent with results previously reported by Simmons et al. (2012), who found all Phillip Island (Victoria) samples tested to be KoRV negative and all samples tested in the Port Macquarie (NSW) area to be KoRV positive. Apart from one sample, standard and real time PCR assays of the samples tested here produced equivalent results. Where presence or absence of KoRV is all that is required for a study, standard PCR can also provide useful data. The small number of diluted positive samples that tested negative by real time PCR indicates that KoRV may not be detected when DNA concentration is quite low. One way to address this may be to screen DNA samples for quantity and quality in order to ascertain the suitability of samples for testing. Additionally, if KoRV detected in scat isolated DNA originates from exogenous infections, the number of cells in which KoRV DNA has integrated may be highly variable, resulting in low copy numbers in some samples. Consequently, until further information is obtained, presence only studies are likely to be most suitable using these methods. In Victorian samples, higher KoRV-A copy number was found in DNA isolated from scats compared to ear tissue, suggesting that koala scats provide a practical source of DNA for detecting the presence of KoRV infections acquired both vertically and horizontally.

Koala conservation research will benefit from the use of an integrated approach, employing the relatively simple method for detecting *C. pecorum* and KoRV described here. Genetic material isolated from koala scats has been shown to provide DNA that can be reliably used to obtain genotypic data, DNA sequence information and to identify gender (Wedrowicz et al. 2013). DNA isolated from scats can provide information on population substructure, relatedness between individuals, gene flow, migration and hybridisation using genotyping (Czarnomska et al. 2013; Piggott et al. 2006; Stenglein et al. 2011). The gender of the individual sampled can be determined by PCR (Wedrowicz et al. 2013), while sequencing of mitochondrial DNA can provide phylogenetic information suitable for evolutionary analyses (Houlden et al. 1999). Genotypic data can also be paired with habitat and environmental models to look for factors that may impede or facilitate gene flow across the landscape using landscape genetics (Storfer et al. 2006).

The ability to detect pathogens in DNA isolated from koala scats may supply critical information that can be combined with genetic data to provide a multidimensional picture of koala biology. Since pathogen DNA sequences can be obtained using this method, non-invasive sampling may also permit studies of pathogen diversity, transmission and spread throughout wild populations. In the case of *C. pecorum*, the likelihood of particular *C. pecorum* strains having high or low virulence could also be investigated using *ompA* markers along with additional *C. pecorum* DNA markers such as *incA* and ORF663 (Mohamad et al. 2014). A general trend of increased *C. pecorum* shedding has been shown for individuals with increasing severity of chlamydial disease (Wan et al. 2011) and there is also evidence of differing infectious load between *C. pecorum* strains (Legione et al. 2016). It may, therefore, also be useful to differentiate individuals with high or low loads of *C. pecorum* using DNA isolated from scats. However, further comparisons of copy number between swabs and scats would be required in order to more accurately gauge the variability of *C. pecorum* copy number in scat isolated DNA and the number of scats needed to give a reliable estimate of infectious load. Pathogen detection from scats could be paired with ecological studies, thereby revealing whether infection might have an effect on behaviour or movement. This information has the potential to provide increased knowledge of many aspects of *C. pecorum* and KoRV infection in koala populations including transmission, pathogenicity and host effects. Together, the plethora of information that is able to be gained from a single koala scat can be used to promote a holistic approach to koala conservation and management.

Acknowledgments Grand Ridge Plantations Pty. Ltd (HVP Plantations) and the Holsworth Wildlife Research Endowment-Equity

Trustees Charitable Foundation are thanked for funding this study. Richard Appleton (HVP) and Chris Allen (NSW Office of Environment and Heritage) are thanked for their support and assistance with the collection of scat samples in the Strzelecki Ranges. David Phalen is thanked for supplying swab isolated DNA for sequencing and quantitative PCR. David Phalen, Damien Higgins and Andrea Casteriano from the Koala Health Hub at Sydney University are thanked for their assistance and for conducting independent analyses. We thank Colleen Wood and the Southern Ash Wildlife Shelter for supplying tissue samples. Cheyne Flanagan and the Port Macquarie Koala Hospital are thanked for the collection of scat samples. Ashley Reed and the Koala Conservation Centre are also thanked for access to and assistance with collecting scats. We also sincerely thank the two anonymous reviewers for their comments and suggestions for improving this manuscript.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval A research permit was obtained under provisions of the Wildlife Act 1975 and National Parks Acts 1975 from the Department of Sustainability and Environment (Permit No. 10004020). Research was also approved by the Monash University Biological Sciences Animal Ethics Committee (AEC No. GIPP/2011/03).

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